

Proposal to form a SETI ComPAS (Communication and Post-Detection Analytics Strategies) SubCommittee within the IAA SETI Permanent Committee

To: Director-General of the International Academy of Astronautics, IAA SETI Permanent Committee Executive, and IAA SETI Permanent Committee

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1. Summary

The SETI Permanent Committee of the International Academy of Astronautics (IAA) is a respected group of experts with specialist knowledge from many disciplines relevant to SETI. The Committee itself is evolving, and SETI science is rapidly changing and expanding, and so are the larger scientific and social landscapes. SETI has always had to contend with key questions, such as: How can signals be detected and confirmed? How should SETI scientists handle the scientific, political, and ethical responsibilities of a detection event? What should happen next? These questions have grown considerably more complex over time and require renewed and intensified efforts to address them comprehensively.

The SETI Permanent Committee has traditionally had a Post Detection Task Force or similar subcommittee. Given that there are at present no such subcommittees, we propose a new SETI Post Detection subcommittee: a proposed title being ComPAS (Communication and Post-Detection Analytics Strategies), which will create a forward-looking strategy for the decade ahead. Such an active presence will also serve as a positive organisational capability for public engagement. *Note: For the purposes of this proposal, we will refer to the proposed subcommittee as entitled ComPas; however, this too is subject to agreement.*

Ever since the formation of SETI, a Post-Detection initiative and capability has been required but often neglected. This capability will be necessary until SETI no longer exists. Arguably, a Post-Detection capability could and should exist outside of the Search, as we may give up on this (in the future) but 'they' may not give up on us; thus, we may have contact when no longer actively searching. We therefore submit that such a subcommittee must 'live' as long as (and alongside) the SETI committee. In that way, Post-Detection expertise will have maintained and developed capabilities, so strategies will be in place, for action.

2. Rationale for Subcommittee Creation

The proliferation of exoplanet discoveries has increased societal awareness that humans may detect extraterrestrial life, including intelligent life. Global interest in the discovery of extraterrestrial intelligence has also intensified through new revelations in astrobiology, high profile technosignature searches, and increased computing and analytical capacity.

The IAA SETI Committee, established in 1974, is unique in that it remains the sole international long-term non-governmental body charged with examining all aspects of the search for extraterrestrial intelligence. This includes an expectation that the IAA SETI Committee will lead discussions of the implications of detecting extraterrestrial intelligent life beyond earth.

During the 1980s the SETI Committee drafted “Declaration of Principles Concerning Activities Following the Detection of Extraterrestrial Intelligence”. The document was approved in 1989 by the IAA Board of Trustees and by the Board of Trustees of the International Institute of Space Law. Although the Declaration and an accompanying position paper was presented to the United Nations Committee on the Peaceful Uses of Outer Space (COPUOS) in 2000, there remains no formal agreement among Earth’s nations in advance of a detection of extraterrestrial life.

The 2010 Declaration of Principles – unanimously adopted by the IAA SETI Committee in Prague in 2010, intended to supersede the Protocols – notes:

http://resources.iaaseti.org/protocols_rev2010.pdf “#7 Post Detection.: *A Post-Detection Task Group under the auspices of the IAA SETI Permanent Study Group has been established to assist in matters that may arise in the event of a confirmed signal, and to support the scientific and public analysis by offering guidance, interpretation, and discussion of the wider implications of the detection.*” By not having a Post Detection Task Group since 2015, the IAA SETI Committee has been failing to follow its own Principles.

The detection of extraterrestrial intelligent life or its technology is anticipated to be one of the most important events in human history. The IAA SETI Committee is an established international scientific body with the collective ability to draw upon appropriate expertise to address both the technological and social impacts of a detection confirmation.

3. The Road Ahead - Terms of reference

SETI has always had key practical dilemmas, Amongst them are:

- How can evidence of ETI be found and confirmed?
- How can we understand the signal?
- Does it contain information; if so, what type and is it interpretable?
- How should that information be shared with the scientific community?
- How should that information be shared with the global public?

These key problems can be provisionally solved, but they cannot have *long-term* solutions, because SETI itself changes continually, so does the larger scientific landscape, and so does the world around us. Furthermore, the rate of change is accelerating; therefore, we must renew and redouble our efforts to keep SETI well-situated.

Aim

In this pursuit, a core aim will be to design and develop our technologies, alongside strategies and agreed philosophies, into a robust and holistic time independent methodology that will establish a framework capable of adaption and evolution, as new knowledge and capabilities are acquired.

Objectives

To investigate, report on, and contribute towards:

- How we can create and maintain an efficient ‘handshake’ between Detection and Post-Detection phases? e.g. discovered data handover; Rio Scale assessments to post detection analytics [DISC Quotient framework] and dissemination;

- How is a discovery process likely to unfold now? (e.g. a long delay between detection of an exoplanet biosphere and follow-up studies; lengthy periods of contention or uncertainty);
- What methods, technologies and strategies do we have for post-detection analytics?
- How should SETI and SETI Area Studies be communicated to the larger science community and to the interested international public, given the global communications and social media landscape?
- What are the *current* global concerns about societal impacts of potential or definite detections of, or contact with, extraterrestrial intelligence?
- What Earth-based studies concerning the evolution of intelligence, social complexity, communication, language, and technology are needed to address questions which will arise upon the detection of LITU (Life in the Universe) or ETI (ExtraTerrestrial Intelligence)?
- Develop a methodology for Post-Detection Analytics and Dissemination;
- Establish a flexible and adaptable SETI Post-Detection Framework.

These questions and action frameworks require our focus. In addition to the expertise within the SETI Permanent Committee, many researchers worldwide are doing relevant work on potential types of detections of extraterrestrial intelligence (including but not limited to classic SETI), policy, the evolution and nature of intelligence and communication and social complexity, scientists' communication with the public, potential social impact of discoveries or contact, and modes of analysis for potential incoming signals. This committee will facilitate conversations with such researchers and research centers.

This ComPAS subcommittee will not address the questions and debates around *de novo* transmissions from Earth.

4. Proposed Initial Structure

We propose a small steering committee primarily composed of longstanding IAA SETI Permanent Committee members with relevant knowledge and extensive leadership experience, who will undertake the task of creating a new, inclusive ComPAS subcommittee and framework which allows contributions from all interested IAA SETI Committee members, including newcomers.

Initial Steering Group Members:

This steering committee will initially comprise of this proposal's signatories: Kathryn Denning, Denise Herzing, Lori Walton, John Elliott and Paul Davies.

From this, a sub-committee comprising relevant expertise, will be created by invitation. A process that will formally commence soon after the committee's formation.

The final SETI ComPAS subcommittee structure, to be announced in summer 2019, will depend on which IAA SETI members wish to be involved. We anticipate that the final SETI ComPAS subcommittee will have Leads for:

- the group's organization/coordination and resource library
- research areas / subject specialists: information sciences, social sciences, biological sciences, astrobiology, astronomy, humanities, law and metalaw, space policy, and communications scholarship concerning social media and news

- next generation capacity-building (i.e. encouraging universities to offer relevant training, encouraging post-graduate students to develop relevant expertise, and facilitating cross-training for professionals to learn about each others' fields)
- liaison with related research centers worldwide

5. Proposed Initial Schedule for SETI ComPAS Subcommittee – or ComPAS Temporary Working Group

Summer 2018 – Summer 2019:

- We will consult within the IAA SETI committee and the larger interested community, to compile information, identify gaps and prepare a draft framework for future work, for a subcommittee which is sustainable over the long term.
- We will consult widely about areas within ComPAS which require discussion, research, and knowledge mobilization.
- We will work together with committee members who have been revising the Rio Scale and the Protocols.
- We will offer briefing documents/presentations about the history and current landscape of post-detection/ComPAS concerns for new/junior SETI Permanent Committee members, so they can rapidly become fully informed participants in these complex conversations.

Details will depend upon resources which can be mustered (e.g. for website hosting, teleconferencing, in-person workshops, etc.), but the aim is to be inclusive, including people who cannot regularly travel, attend expensive overseas meetings, etc.

Committee Meeting at IAC 2019, Washington DC, 21-25 October 2019:

- We will present a new framework / strategy document for this SETI ComPAS Subcommittee. We will provide a written paper for the committee members who are unable to attend the IA Congress.

Special Note: This new subcommittee will be open to any interested IAA SETI Permanent Committee members who have relevant expertise. We invite those who are interested to identify themselves, so we can start building a common conversation which will take us through the next decade. Please send an email to John Elliott at setijon@aol.com and Kathryn Denning at kdenning@yorku.ca with a paragraph about your pertinent interests, a list of any specifically relevant publications and/or current research or policy endeavours, and a summary of your IAA SETI membership history and prior activities (if any) within the IAA SETI group and within the larger IAA. With this information, we can start creating a mutually aware network of participants. We intend to regularly report to and seek input from the larger IAA SETI Committee.